

TEACHER RESOURCE | VERSION 1.0

The TRACE Teacher Pack

Plan AI-supported learning without handing over the thinking. A compact planning and reflection kit for classroom teachers.

T
Target the thinking

R
Retrieve before AI

A
Ask with boundaries

C
Check the output

E
Edit and explain

Honest label: TRACE is Chad's interpretation and working framework, not a validated intervention. AI can question, prompt, and reflect. The learner still has to supply the thinking worth reflecting on.

Two connected ways to use TRACE

FOR STUDENT LEARNING	FOR A BOUNDED AI JOB
Try first Reveal reasoning Annotate Check Evolve	Task Relevant context Actions and aids Checks Exit and escalation
The learner attempts, explains, inspects, and revises.	The teacher defines the work, evidence, limits, validation, and handoff.

The line in the sand: strong performance while assistance is available is not the same as learning that remains when assistance is removed. The wrapper around AI matters because it changes what the student still has to do.

START HERE
1. Read the completed example on page 2. 2. Plan one bounded job on page 3. 3. Give students the revision handout on page 4.

WORKED CLASSROOM EXAMPLE

From a vague AI activity to a bounded learning task

Teacher-created practice question: Evaluate whether schools should use generative AI to support student feedback.

T - Target	Make a defensible judgment using a claim, relevant evidence, a counterclaim, and a reasoned conclusion.
R - Retrieve	Write for five minutes from memory before any AI support. Mark the sentence where the judgment becomes visible.
A - Ask	AI may ask diagnostic questions only. It may not rewrite, score, add facts, or provide a model answer.
C - Check	Compare the response with teacher criteria and approved class sources. AI comments remain suggestions, not authority.
E - Edit	Revise one paragraph, then write: 'I changed... because... I still need to check...'

Bounded prompt
You are a questioning coach. Read this anonymous practice paragraph. Ask no more than three questions about the claim-evidence connection, counterclaim, and final judgment. Do not rewrite the paragraph, add facts, score it, or provide a model answer. Stop after the questions.

What a useful result looks like

AI MAY RETURN	THE STUDENT MUST STILL DO
Three diagnostic questions about missing reasoning or weak connections.	Choose which question matters, check sources and criteria, revise the paragraph, and explain the change.

Safety preflight: use an anonymous practice response; never paste student names or private records; never treat AI scoring as authoritative; a teacher remains responsible for the task, evidence, and final judgment.

TEACHER PLANNING SHEET

Build one bounded AI job

Start with the learning, not the model. Complete this sheet before a student or agent receives the prompt.

1. LEARNING TARGET What thinking must remain with the learner?	2. STARTING EVIDENCE What must happen before AI enters the task?
3. ALLOWED AI ROLE Questioner, critic, formatter, tutor - or no AI at all?	4. FORBIDDEN ACTIONS What must it never write, decide, score, publish, or change?
5. PERMITTED CONTEXT What approved, privacy-safe material may it read?	6. EXACT DELIVERABLE Name the format, maximum length, and destination.
7. DETERMINISTIC CHECKS Which source, criteria, test, or human judgment verifies it?	8. EXIT AND ESCALATION When must the agent stop and return control to a human?

FINAL CHECK ☐ No identifiable student information ☐ Genuine attempt before assistance ☐ No autonomous grading or publishing ☐ Human review is named

STUDENT HANDOUT

TRACE your revision

Name: _____ Task: _____

QUICK EXAMPLE - WHAT AM I SUPPOSED TO DO?

My first attempt: 'AI feedback is always fair because computers are not biased.'

My bounded prompt: Ask me up to three questions about my evidence, counterclaim, and judgment. Do not rewrite my answer.

My reflection: 'I changed always fair because I had not supported it and AI systems can reflect bias. I still need to check an approved source.'

☐ I attempted the task before asking AI.

☐ I can explain the idea without reading the AI response.

☐ I checked the response against class criteria or an approved source.

☐ I identified one useful suggestion and one possible weakness.

☐ I revised in my own words and can explain what I changed.

☐ I did not share names, private records, or sensitive information.

MOST USEFUL QUESTION The most useful question or suggestion was... _____ _____	JUDGMENT I did not accept this part because... _____ _____
REVISION I changed... because... _____ _____	UNCERTAINTY I still need to check or ask... _____ _____

WHY THIS IS DESIGNED THIS WAY

Retrieval, informative feedback, prompted self-explanation, and structured self-assessment support the rationale for an attempt-inspect-revise loop. They do not prove that TRACE or Chad's applications improve IB results.

Roediger & Karpicke (2006), doi.org/10.1111/j.1467-9280.2006.01693.x | Agarwal, Nunes & Blunt (2021), doi.org/10.1007/s10648-021-09595-9 | Wisniewski, Zierer & Hattie (2020), doi.org/10.3389/fpsyg.2019.03087 | Bisra et al. (2018), doi.org/10.1007/s10648-018-9434-x | Bastani et al. (2025), doi.org/10.1073/pnas.2422633122

TRY THE TOOLS

Exam Practice Studio: macchady-spec.github.io/ib_paperPractice/ibdsdiag.html

Response Studio: macchady-spec.github.io/ib_paperPractice/

Help me make this better. Try TRACE once, then send one useful observation to chad.aigroove@gmail.com. Tell me where it helped - or where it fell apart. Real classroom feedback is more useful than applause.